

# Matsuyama in Space

Agricultural productivity, connectivity and structural transformation in Korea, 1965–85

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[Conference] · Work in progress — preliminary, please do not cite

[Affiliations]

# Motivation

Korea, 1965–85: one of the fastest structural transformations on record.

|                               | ~1960/65                | ~1980/90                |
|-------------------------------|-------------------------|-------------------------|
| Real GDP per capita           | 1                       | ×4 (1985) <sup>†</sup>  |
| Agriculture, employment share | 59% (1965) <sup>†</sup> | 25% (1985) <sup>†</sup> |
| Manufacturing, VA share       | 15% (1965)              | 30% (1985)              |
| Urban population share        | 28% (1960)              | 57% (1980), 74% (1990)  |
| Farm household population     | ~14m                    | <6m (1990)              |

Sources: World Bank WDI; national statistics. <sup>†</sup> approximate, being updated from primary sources.

- Poverty was concentrated in the **countryside**. The transformation is a story about where farm labour went, and why.
- At the same time: industrial parks, an expressway network, irrigation, high-yield rice, and a closed, policy-priced rice market.
- Standard question: does farm productivity growth *push* labour into industry, or *anchor* it on the farm?

## Question

When agricultural productivity rises, does a place **commercialise** its farming or **release** farm labour into non-farm work and migration — and how does the answer depend on its connections to crop markets and to non-farm jobs?

## What we do

1. A **spatial general-equilibrium model** of structural transformation: non-homothetic demand, migration nested with occupation choice, protected rice, import-competing grains, labour-saving mechanisation.
2. A **baseline that reproduces history**: calibrated to Korea's 1965–85 transformation, with later urbanisation generated by the model.
3. A **township panel** (1960–90, annual; being digitised) and quasi-experimental designs (reservoirs, road paving) to discipline the mechanism.

## Preview of results (preliminary)

- The sign of the farm-employment response to a farm productivity shock is governed by two distinct access margins, which connectivity moves *together*.
- The baseline reproduces fourfold growth, a 38-point fall in farm employment and urbanisation from 32% to 63% through Engel effects, grain import competition and migration.
- Occupation wedges generate an agricultural productivity gap that **widens** during the transformation.
- Companion reduced-form evidence: the same road-access shock releases labour near industrial parks and retains it in remote townships.

### Contribution

Separately identify farm productivity shocks, industrial labour-demand shocks and connectivity, and show where their interaction produces commercialisation versus labour release — then aggregate it in general equilibrium.

## Related literature

- **Agricultural productivity and industrialisation:** Matsuyama (1992); Gollin, Parente & Rogerson (2007); Gollin & Rogerson (2014); Uy, Yi & Zhang (2013); Teignier (2018) — *we make the open/closed distinction spatial and township-specific*.
- **Spatial structural change and growth miracles:** Eckert & Peters; Budí-Ors & Pijoan-Mas; Fan, Peters & Zilibotti (2023); Tombe & Zhu (2019); Hao, Sun, Tombe & Zhang (2020).
- **Agricultural technology and labour:** Foster & Rosenzweig (2004); Bustos, Caprettini & Ponticelli (2016); Hayami & Ruttan (1971); Cheung & Yang (2024) — *we separate land-augmenting from labour-saving change, and crop-market from labour-market access*.
- **Agricultural productivity gap and sectoral frictions:** Gollin, Lagakos & Waugh (2014); Lagakos & Waugh (2013); Hsieh, Hurst, Jones & Klenow (2019); Lagakos, Mobarak & Waugh (2023).
- **Transport, market access and agriculture:** Donaldson (2018); Sotelo (2020); Allen & Arkolakis (2014); Redding & Rossi-Hansberg (2017).
- **Micro to macro:** Nakamura & Steinsson (2018); Adão, Arkolakis & Esposito (2019); Buera, Kaboski & Townsend (2023).

## Setting and motivating evidence

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# Korea's spatial big push

## Industry

- Heavy and Chemical Industry declaration (1973): steel, non-ferrous metals, machinery, shipbuilding, electronics, chemicals.
- Delivered through industrial parks: Guro (1967), Pohang (1968), Changwon and Onsan (1974), the Gyeongbu corridor.

## Roads

- Gyeongbu Expressway (Seoul–Busan) opened 1970; five more motorways by 1981.
- Highway coverage roughly doubled 1970–80; paved local road density roughly tripled 1967–80.

## Agriculture

- Post-1950 land reform: small owner-operated farms under a 3-ha ceiling.
- Reservoirs and irrigation; Tongil high-yield rice released 1972.
- From 1969: rice import controls and government procurement (dual grain-price policy).

### The key institutional fact

The rice market was **closed to the world** and policy-priced, but **integrated within Korea**. A township was open to the nation while the nation was closed. Whether a place is “open” or “closed” is a township-level question, and roads change the answer.

# Motivating evidence: connectivity cuts both ways

Companion reduced-form work: township first differences, 1970/75–1990, on growth in road-network market access (MA), instrumenting away nearby destinations;  $\sim 1,300$  townships; SE clustered by county.

| Effect of $\Delta \log$ MA on     | coef. (s.e.) |
|-----------------------------------|--------------|
| farm VA per worker                | 0.70 (0.30)  |
| farm labour intensity             | −0.61 (0.29) |
| non-farm pop., nearest IP decile  | −1.81 (0.80) |
| non-farm pop., farthest IP decile | +0.63 (0.31) |

IP = industrial park. Preliminary.

- Access induces **mechanisation**: more output per worker, less labour per unit of land.
- The population response **reverses sign** around the median distance to an industrial park: release near jobs, retention far away.
- New service establishments respond in middle-to-remote townships, not near parks.

⇒ The same connectivity shock can release or retain labour. We need a model that says *where* and *why*, and aggregates it.

# Mechanism

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# Matsuyama (1992), made spatial

Matsuyama (1992): manufacturing has learning-by-doing.

- **Closed economy.** Farm productivity  $\uparrow \Rightarrow$  food needs met with fewer workers  $\Rightarrow$  labour **released** to industry  $\Rightarrow$  faster growth (the “food problem”).
- **Open economy.** Farm productivity  $\uparrow \Rightarrow$  comparative advantage in farming  $\Rightarrow$  labour **retained**  $\Rightarrow$  slower industrialisation.

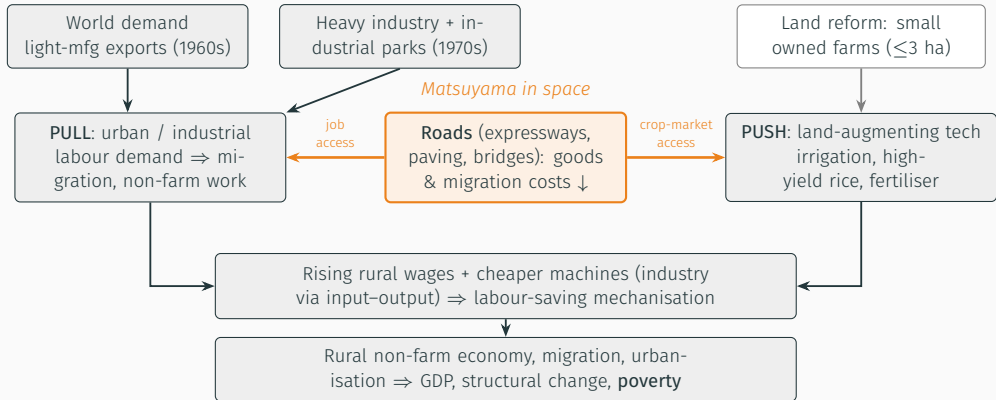
The regime is usually national.

## Our hypothesis

Agricultural productivity growth induces **commercialisation** where access to crop demand dominates, and **labour release** where access to non-farm opportunities dominates.

In Korea each township sits somewhere between the two regimes, and roads, parks and rural programmes move it.

# The mechanism chain

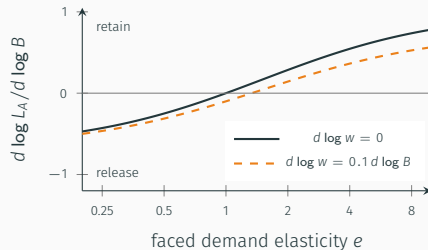


# A township in one equation

Competitive farms,  $Q_i = (B_i T_i)^\alpha L_{Ai}^{1-\alpha}$ ; land-augmenting shock  $B_i$  (e.g. a reservoir). Local crop demand faced with elasticity  $e_i$ ; opportunity wage  $w_i$ .

$$d \log L_{Ai} = \frac{\overbrace{\alpha (e_i - 1) d \log B_i}^{\text{commercialisation vs. food problem}} - \overbrace{e_i d \log w_i}^{\text{labour release}}}{\alpha e_i + 1 - \alpha}$$

- **Crop-market access** raises  $e_i$ : isolated  $\approx$  local food demand elasticity ( $< 1$ ); connected  $\rightarrow$  trade elasticity  $\theta$  ( $\gg 1$ ).
- **Job access** moves  $w_i$ : industrial pull, migration.
- Fixed  $w$ : labour retained iff  $e_i > 1$  — Matsuyama's two regimes as two ends of the access distribution.
- Connected places also respond *more* to the opportunity wage.



Illustrative,  $\alpha = 0.4$ ; derivation in backup.

# What the sign can and cannot tell us

A tempting test: “the effect of irrigation on farm employment flips sign with market access”. Three reasons it is not, by itself, informative:

1. **Gravity builds in the flip.** With CES trade, connected places face  $e = \theta > 1$  and isolated places  $e < 1$ . The data identify **where** on the access distribution the crossing occurs — which disciplines trade costs and Engel curves.
2. **Connectivity moves both terms.** Better-connected townships also have non-farm jobs, cheaper machines and lower migration costs, which push towards release. We need **separate** measures of crop-market and job access.
3. **Irrigation is not Hicks-neutral.** Paddy irrigation and high-yield rice raised labour per hectare (double cropping, transplanting). The factor bias must be estimated jointly.

## Our approach

The model, with trade, migration and Engel parameters estimated from other moments, **predicts** the interaction coefficient; the township event studies estimate it. A held-out test. Township retention is also not aggregate retardation, so the aggregate consequences come from general equilibrium.

# Labour-saving technology: mechanisation

A tiller cuts labour per hectare by a fraction  $\chi$  and costs  $u_t = p_{m,t}(r_t + d)$  a year. A farm of size  $\ell$  adopts iff

$$\chi \lambda_0 \ell w_{iA,t} \geq u_t \iff \ell \geq \ell_{it}^* = \frac{p_{m,t}(r_t + d)}{\chi \lambda_0 w_{iA,t}}, \quad \text{adoption}_{it} = 1 - F_i(\ell_{it}^*).$$

- $F_i$ : observed landholding distribution, a legacy of land reform.
- Adoption responds to farm wages (**industrial pull**), machine prices (**domestic machinery industry** via input-output), and farm size.
- Induced innovation (Hayami–Ruttan) with a micro-founded extensive margin.

## Why it matters

Labour-saving change **releases** labour; land-augmenting change can **absorb** it (Bustos–Caprettini–Ponticelli 2016). Korea had both, in different places and at different times.

# Model

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Sequence of spatial equilibria; state = lagged population.

- Locations  $d \in \mathcal{N}$ ; current quantification: five (Seoul, Busan, Changwon, Daegu, rural Korea).
  - **Four sectors:**
    - **Rice:** protected, internationally non-traded; can be mechanised.
    - **Other agriculture:** competes with imports at world prices.
    - **Manufacturing:** traded; the export sector.
    - **Services:** non-traded.
  - Input-output linkages; trade costs between locations; sector-specific agglomeration.
- Non-homothetic (PIGL) demand.
  - Migration across locations, nested with occupation choice within them.
  - Heterogeneous producers; farms select into mechanisation.
  - Local land ownership; government spending buys output.
  - Small open economy; trade balance pinned to an exogenous path.
  - Growth: sectoral TFP growth and rising foreign demand for manufactures.

## Preferences: the food problem

PIGL indirect utility (Boppart 2014) at income  $y$  and prices  $P_{dj}$ ,  $\mathcal{P}_d = \prod_j P_{dj}^{\alpha_j}$ :

$$V_d = \frac{1}{\eta} \left( \frac{y}{\mathcal{P}_d} \right)^\eta - \sum_j v_j \log P_{dj}, \quad \psi_{dj} = \alpha_j + v_j \left( \frac{y}{\mathcal{P}_d} \right)^{-\eta}, \quad \sum_j \alpha_j = 1, \sum_j v_j = 0.$$

- $\eta > 0$ ,  $v_{\text{food}} > 0$ : the food share falls with real income; the services share rises.
- Aggregates exactly over finitely many income groups: farm and non-farm households demand different baskets, so the *composition* of local income shapes local demand (rural services).
- Migration and welfare use the money-metric equivalent income  $\mathcal{Y} = \mathcal{P}_0[\eta(V + z_0)]^{1/\eta}$ , so choices and welfare rank places exactly as utility does.

## Location $\times$ occupation choice

Agent in origin  $o$  chooses location  $d$  and occupation  $\kappa \in \{\text{farm, manufacturing, services}\}$  to maximise  $W_{od} + \log b_{d\kappa} + \log \mathcal{Y}_{d\kappa} + v_{i,d\kappa}$ , with nested-logit (GEV) shocks:

$$\pi_{\kappa|d} = \frac{(b_{d\kappa} \mathcal{Y}_{d\kappa})^\varepsilon}{\sum_{\kappa'} (b_{d\kappa'} \mathcal{Y}_{d\kappa'})^\varepsilon}, \quad \Phi_d = \left[ \sum_{\kappa} (b_{d\kappa} \mathcal{Y}_{d\kappa})^\varepsilon \right]^{1/\varepsilon}, \quad \mu_{od} \propto \left( \frac{\bar{V}_d L_{d,t-1}^\ell \Phi_d}{\delta_{od}} \right)^\nu.$$

- $\nu \leq \varepsilon$ : switching sector within a place is easier than moving.
- $b_{d\kappa}$ : non-pecuniary value of an occupation — attachment to farming, barriers to leaving it.
- Each location–occupation labour market clears at its own wage.
- Welfare by origin is the inclusive value;  $\partial \mathbb{E}U_o / \partial \log \mathcal{Y}_{d\kappa} = \mu_{od} \pi_{\kappa|d}$ .
- Limit  $\varepsilon \rightarrow \infty$ ,  $b = 1$ : one integrated wage per location.

# Why sectoral wages: the agricultural productivity gap

With one wage per location, farm and non-farm value added per worker differ only through labour shares  $\lambda^k$ :

$$\text{APG}_d \equiv \frac{\text{VA}_d^A/L_d^A}{\text{VA}_d^N/L_d^N} = \frac{w_d^A/\lambda_d^A}{w_d^N/\lambda_d^N} \stackrel{w^A \equiv w^N}{=} \frac{\lambda_d^N}{\lambda_d^A} \gtrsim 1,$$

far from the gaps measured in poor economies (Gollin–Lagakos–Waugh 2014).

## Wage-gap decomposition

$$\log \frac{\mathcal{Y}_{dA}}{\mathcal{Y}_{dN}} = \underbrace{\frac{1}{\varepsilon} \log \frac{L_{dA}}{L_{dN}}}_{\text{finite supply elasticity}} - \underbrace{\log \frac{b_{dA}}{b_{dN}}}_{\text{non-pecuniary wedge}}$$

- A large farm share at a *lower* farm wage requires  $b_A > b_N$ .
- When farm labour demand falls, finite  $\varepsilon$  keeps workers in farming at a falling relative wage: the gap **widens** before switching and migration catch up.

# Production, trade and equilibrium

- **Producers:** monopolistic competition with productivity draws; entry, exporting and (in rice) mechanisation are a nested discrete choice — only active producers mechanise or export.
- **Mechanisation** substitutes machinery for labour. Its profit gain

$$\Delta = \left( \xi \frac{C}{\bar{C}} \right)^{\sigma-1} - 1$$

risks with  $w/P_{\text{machinery}}$ : adoption turns on as **wages rise and machines get cheaper**.

- **Trade:** CES across origins with iceberg costs; foreign varieties at world prices; aggregate export demand elasticity  $\sigma_x$ . External balance  $EX - IM = NX_t^*$  pins the domestic price level.
- **Equilibrium:** migration and occupation shares, PIGL demand, firm choices, goods, labour and land market clearing, government budget, external balance. Solved as a fixed point; adding-up holds to machine precision.
- **Initial condition:** 1965 amenities inverted so the observed 1965 population is an equilibrium; urbanisation after 1965 is a **model outcome**.

## Quantification (preliminary)

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# Calibrating to history: 8 parameters, 8 targets

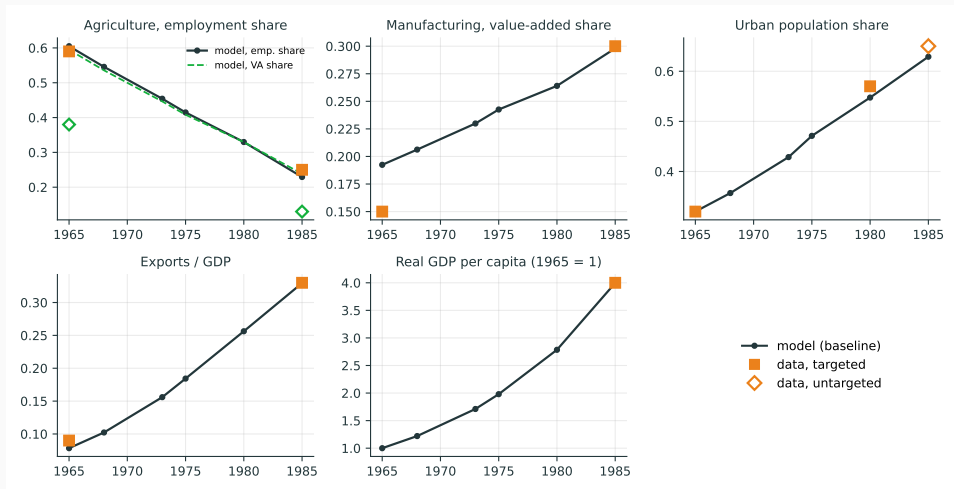
| Parameter                           | Value   | Main target            |
|-------------------------------------|---------|------------------------|
| Manufacturing TFP growth            | 4.1%/yr | real GDP p.c., 1985/65 |
| Farm and services TFP growth        | 3.2%/yr | mfg VA share 1985      |
| PIGL income elasticity $\eta$       | 0.39    | farm emp. share 1985   |
| PIGL food shifter $v_{\text{food}}$ | 0.51    | farm emp. share 1965   |
| Mfg weight in non-food demand       | 0.02    | mfg VA share 1965      |
| Foreign demand, 1965 level          | —       | exports/GDP 1965       |
| Foreign demand growth               | 24%/yr  | exports/GDP 1985       |
| Urban non-farm TFP premium          | 1.17    | urban share 1980       |

Exactly identified; the 1965 amenity inversion is redone at every evaluation.

**Set externally:**  $\sigma = 4$ , Pareto shape 5, migration elasticity  $\nu = 5$ , mechanisation gain  $\xi = 1.2$ , land shares, input-output coefficients, trade and migration costs.

**Current simplifications:** five stylised regions; six periods (1965–85); rice half of food demand; stylised input-output table.

# The baseline reproduces Korea's transformation



Green: farm value-added share (model dashed; data hollow). Several data targets are approximate and being updated from primary sources.

# What drives it, and where it misses

## Accounting for the transformation

- Farm employment falls 38 points ( $0.61 \rightarrow 0.23$ ) through **Engel effects** on rising income, **grain import competition** (import share of other farm demand  $0.10 \rightarrow 0.76$ ) and **migration**.
- Urbanisation  $0.32 \rightarrow 0.63$  is generated, not imposed (1985 untargeted:  $0.63$  vs  $0.65$ ).
- Export share  $8\% \rightarrow 33\%$  requires foreign demand for manufactures growing  $\sim 24\%/yr$ .
- Rice mechanisation takes off only after the mid-1970s, as wages rise.

## Where it misses

- **No productivity gap** with integrated local labour markets: 1965 farm VA share  $0.60$  vs  $\approx 0.38$  in the data  $\Rightarrow$  occupation wedges (next slide).
- 1965 manufacturing share too high ( $0.19$  vs  $0.15$ ): stylised input-output structure  $\Rightarrow$  Korean IO tables.
- No relative-price (Baumol) channel in demand.

# Occupation wedges open — and widen — the productivity gap

|                                       | 1965      |           |       | 1985      |           |       |
|---------------------------------------|-----------|-----------|-------|-----------|-----------|-------|
|                                       | farm emp. | $w_A/w_N$ | APG   | farm emp. | $w_A/w_N$ | APG   |
| integrated ( $\varepsilon = \infty$ ) | 0.605     | 0.827     | 0.958 | 0.229     | 0.868     | 1.053 |
| $\varepsilon = 5, b_A = 1$            | 0.530     | 1.051     | 1.214 | 0.218     | 0.908     | 1.107 |
| $\varepsilon = 5, b_A = 1.5$          | 0.627     | 0.802     | 0.930 | 0.314     | 0.676     | 0.803 |
| $\varepsilon = 5, b_A = 2$            | 0.680     | 0.653     | 0.758 | 0.399     | 0.545     | 0.638 |

Comparative statics: 1965 amenities re-inverted per case, other parameters at the integrated calibration (not yet recalibrated). APG = farm VA per worker / non-farm.

1.  $b_A = 1$  is *not* frictionless: equal pay would mean equal occupation shares, so a large farm share needs a higher farm wage.
2. With a farm wedge the gap **appears and widens** during the transformation (APG 0.76  $\rightarrow$  0.64): falling farm labour demand lowers the farm wage faster than workers leave.
3. The wedge also **slows** the transformation. Separating  $b_A$ ,  $\varepsilon$  and the Engel parameters requires farm and non-farm earnings data.

## Data and empirical strategy

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## A new township panel, 1960–90

Annual township-level data being digitised from statistical yearbooks:

| Block             | Variables                                                                                                      | Role in the model                                  |
|-------------------|----------------------------------------------------------------------------------------------------------------|----------------------------------------------------|
| Population        | population; employed / unemployed / inactive; employment by sector                                             | amenities, occupation shares, non-employment       |
| Agriculture       | land by type (paddy / dry); farm households by landholding size; reservoirs (capacity, date); crops, livestock | farm-size distribution; irrigation shock; crop mix |
| Infrastructure    | roads paved / unpaved; bridges; ports; electricity                                                             | local access; dated shocks                         |
| Municipal finance | revenue, central transfers, spending by function                                                               | local public investment                            |
| Schools           | schools, students, teachers, enrolment                                                                         | human capital                                      |

Complemented by county manufacturing surveys, the digitised road network (1967/75/87), and national accounts.

# Testing the mechanism: reservoirs

- Event studies of reservoir completions, heterogeneous by **predetermined** access to *crop markets* and, separately, to *non-farm employment*.
- Treatment follows the irrigation *command area*, not the township containing the dam.
- Placebo: dry-field townships.
- Outcome: farm employment *share* (migration contaminates levels).
- A reversal requires the effect to **cross zero** within the observed support of access.

| Channel                 | Evidence                                                            |
|-------------------------|---------------------------------------------------------------------|
| Commercialisation       | marketed output, farm-gate prices, farm labour demand, land returns |
| Labour release          | labour per hectare ↓, mechanisation, non-farm work, migration       |
| Local demand spillovers | rural services after farm-income gains                              |

Model side: the predicted interaction coefficient, with trade, migration and Engel parameters set from other moments.

# From township estimates to aggregate effects

Township designs identify *relative* effects; year effects absorb national prices, migration to cities and fiscal costs. The model supplies the general-equilibrium **intercept**.

| Parameter                                     | Identified from                                                               |
|-----------------------------------------------|-------------------------------------------------------------------------------|
| Trade elasticity, distance cost               | price gaps across markets; market-access event studies                        |
| Migration elasticity and costs                | origin–destination flows; expressway interchange openings                     |
| Occupation elasticity $\epsilon$ , wedges $b$ | sectoral employment responses to shift-share industrial demand; earnings gaps |
| Engel parameters                              | household-survey Engel curves                                                 |
| Mechanisation threshold                       | adoption by farm-size class $\times$ wage and machine-price shocks            |
| Irrigation factor bias                        | reservoir event studies; labour per hectare                                   |

**Counterfactuals:** subtract measured shocks from history and decompose growth, structural change and **poverty** across drivers (world demand, industrial policy, roads, rural programmes) with Shapley values; price complementarities such as parks  $\times$  roads.

# Summary and next steps

## Summary

- Whether farm productivity growth retains or releases labour depends on two access margins — to crop markets and to non-farm jobs — which connectivity moves together.
- Korea's closed-but-integrated rice market makes the Matsuyama regime a township-level variable.
- A spatial GE model calibrated to history reproduces the 1965–85 transformation; occupation wedges deliver a widening productivity gap.

## Next

- A township-level spatial model with competitive farms on paddy and dry land, irrigation as a factor-biased shock, and the observed farm-size distribution.
- Reservoir and road-paving event studies on the township panel.
- Recalibrate with sectoral earnings and Korean input-output tables; annual clock; capital and schooling for the aggregate decomposition.

# Thank you

Comments very welcome  
[email]

## Backup: derivation of the township equation

Competitive farm, Cobb–Douglas:  $p(1 - \alpha)Q/L_A = w$  with  $Q = (BT)^\alpha L_A^{1-\alpha}$  gives

$$\log L_A = \frac{1}{\alpha} [\log(1 - \alpha) + \log p + \alpha \log(BT) - \log w].$$

Faced demand  $Q = D p^{-e}$ , so  $d \log p = -\frac{1}{e} d \log Q = -\frac{1}{e} [\alpha d \log B + (1 - \alpha) d \log L_A]$ . Substituting and collecting  $d \log L_A$ :

$$d \log L_A \left( \alpha + \frac{1 - \alpha}{e} \right) = \alpha \left( 1 - \frac{1}{e} \right) d \log B - d \log w \implies d \log L_A = \frac{\alpha(e - 1) d \log B - e d \log w}{\alpha e + 1 - \alpha}.$$

- $e \rightarrow \infty$  (price-taker):  $d \log L_A = d \log B - \frac{1}{\alpha} d \log w$ .
- Partial equilibrium and Hicks-neutral; the quantitative model endogenises  $e$  and  $w$  and allows factor bias.

## Backup: nested entry, adoption and export

With  $x = \phi^{\sigma-1}$ , every option's profit is linear in  $x$ : option  $k$  has slope  $a_k$  and fixed cost  $c_k$ . Rice: exit  $(0, 0)$ , traditional  $(a^T, F)$ , mechanised  $(a^M, F + \check{F})$ , mechanised exporter  $(a^M + a^X, F + \check{F} + \tilde{F})$ . The producer picks the upper envelope:

$$\text{"option } k \text{ or higher"} \iff x \geq X_k, \quad X_k = \min_{k' \geq k} \max_{i < k'} \frac{c_{k'} - c_i}{a_{k'} - a_i}.$$

A traditional segment exists iff

$$\frac{\check{F}}{F} > \Delta, \quad \frac{\check{F} + \tilde{F}}{F} > \Delta + \frac{a^X}{a^T}, \quad \Delta \equiv \left( \xi \frac{c}{\check{c}} \right)^{\sigma-1} - 1.$$

Fixed costs are in units of local labour, so they rise with wages. In the planned township model this block is replaced by competitive farms with the farm-size threshold.

## Backup: decomposition and poverty

**Shapley decomposition** over drivers  $D$  (world demand, capital, human capital, demography, industrial credit, parks, roads, rural programmes, residual TFP):

$$\phi_d = \sum_{S \subseteq D \setminus \{d\}} \frac{|S|! (|D| - |S| - 1)!}{|D|!} [Y(S \cup \{d\}) - Y(S)].$$

Pairwise complementarity:  $Y(a, b) - Y(a) - Y(b) + Y(\emptyset)$ .

**Poverty** from model incomes by (location, occupation, land class) with within-cell lognormal dispersion  $\sigma_c$  from household surveys:

$$H_t = \sum_c \omega_{c,t} \Phi\left(\frac{\ln Z - \ln \bar{y}_{c,t}}{\sigma_c} + \frac{\sigma_c}{2}\right).$$

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